

Question	Answer	Mark	Mark scheme	Additional guidance
22	$b = 3a$	P1	for $a\binom{2}{6} + b\binom{8}{2} = k\binom{13}{6}$ oe or for setting up a linear equation in a or b eg $2a + 8b = 13k$ or $6a + 2b = 6k, k \neq 0, 1$ or for $2a + 8b = 13$ and $6a + 2b = 6$ or for $\frac{2a+8b}{6a+2b} = \frac{13}{6}$ oe	Accept any non zero value substituted for k
		P1	for process to solve the simultaneous equations to get $b = \frac{3k}{2}$ and $a = \frac{k}{2}$ or $b = \frac{3}{2}$ and $a = \frac{1}{2}$ or both $2a + 8b = 13k$ and $6a + 2b = 6k$ with process to eliminate k	
		A1	for $b = 3a$ oe	